Concrete Proof-Rule Cards: V3, V5, V6, and E1

Tactix Development Project

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Executive summary

The interactive proof-rule guide's Substitute, Sweep, Congruence, and BVE entries have been replaced with concrete indexed transitions, ordered proof deltas, reconstruction obligations, and checkable worked examples. Their authoritative entries in `docs/calculus.md` were tightened where the existing prose did not name the emitted objects or their order. This is a documentation-only change; no solver behavior or performance changed.

1 Findings and results

V3 Substitute. The rule now defines the SCC representative map, one transient BIG-path bridge per rewritten polarity, the normalized image of each touched clause, add-before-delete ordering, and the symmetric witness pair used to recover each substituted variable. The contradictory-SCC branch names both RUP units and the final empty-clause chain.

V5 Sweep. The environment now records how root-false literals are stripped, and the two equivalence directions are named separately. The guide distinguishes main-kernel RUP replay from translation of a sub-solver derivation and states the all-or-nothing validation rule. A worked binary example gives the exact hint order for both directions.

V6 Congruence. The single "equivalence ladders" placeholder has been split into the actual proof families: direct AND chains with BIG bridges, the four ITE condition cases, XOR sign-pattern ladders, and degeneracy ladders. The common order—transients, final conclusions, reverse transient deletions—is explicit. The examples show exact duplicate-AND chains and a gate that becomes constant only after an earlier equivalence merge.

E1 BVE. Positive and negative parents, indexed resolvents, root-unit stripping, the fixed two-parent LRAT order, gate-reduced pair scope, and both-sided reconstruction witnesses are now explicit. The examples cover one real resolvent and a tautological pair that emits no resolvent but still needs witnesses.

2 Evidence

The revisions were checked against the corresponding implementations in `src/rules/substitute`, `src/rules/sweep`, `src/rules/congruence`, and `src/rules/bve`. The offline page parses through Lynx and retains exactly 24 rule cards, 24 formal panels, 24 informal panels, and 24 example panels. Citation, section-reference, document-size, comment-hygiene, rule-policy-boundary, and whitespace checks pass. No solver binary was changed, so no solver or performance measurement is attributed to this increment.

3 Risks and decisions

The HTML remains a derived teaching view; `docs/calculus.md` remains authoritative. The expanded cards are necessarily denser, but the existing tabs keep the formal detail separate from the plain-language explanation and examples. Three untouched cards still contain compact placeholders: Vivify's "reason-chain", HyperBinaryRes's optional deletion, and Instantiate's "reason-chain including C". They were left outside this four-rule scope rather than revised without a dedicated semantic audit.

4 Next actions

Review the four cards in a graphical browser. If the level of detail is right, the remaining three shorthand cards can receive the same focused treatment. The solver-development handoff remains unchanged: resume Kissat-default parity with the `SP-COMPACT-1` row.